

Structure-Verified RLVR for Label-Efficient Pathology Instance Segmentation

Working title: *S-Seg-RLVR — Non-Differentiable Structural Verifiers as Rewards for Weakly-Supervised Nuclei and Gland Segmentation*

Target venue: MICCAI 2027 (fallback: ISBI 2027 / MIDL 2027)

Hypothesis

We hypothesize that pixel-overlap objectives (Dice/IoU) are the wrong training signal for pathology instance segmentation, and that the structural quantities pathologists actually care about — instance count, instance separation, and object topology — are the right inductive bias but have been unusable as *losses* because they are discrete and non-differentiable. Reinforcement learning with verifiable rewards removes that constraint: it can score the true discrete quantity directly, so we can supervise segmentation with the exact structural criterion rather than a differentiable surrogate of it, and derive that criterion from weak (point/count) annotation rather than dense masks.

Motivation and gap

Two nearly-touching nuclei merged into one instance barely dents Dice but destroys the count and, downstream, the grade — under-segmentation of clustered nuclei is the canonical clinically-relevant failure. The structural targets that catch this (number of connected components, non-overlap of instances, a gland being a closed epithelial ring around a lumen) are **discrete**. The topology-aware segmentation literature has responded by engineering a decade of *differentiable approximations* — persistent-homology losses (Hu 2019; Clough 2019), soft-skeleton cDice (Shit 2021), induced-matching of persistence barcodes (Stucki 2023) — explicitly because Betti numbers and instance counts cannot be backpropagated. RLVR sidesteps that machinery entirely: GRPO evaluates a scalar reward per rollout and never differentiates it, so the discrete structural quantity *is* the reward.

Positioning against current work:

- **Reasoning-RL for segmentation is natural-image** (Seg-Zero, Seg-R1, LENS). The medical GRPO wave is grounding / VQA / interactive clicking (MedGround-R1, MedReasoner, MedSAM-Agent), not dense pathology instance masks.
- **RL in pathology is stuck on classification/navigation** (RLogist, PEAN, Qaiser & Rajpoot) or on narrow post-processing — the only nuclei-segmentation RL I could find optimizes an adaptive foreground threshold on an already-trained network's probability map, or does superpixel merge/split (stateless actor-critic).

- **Structural priors for nuclei exist only as differentiable losses** (distance-map regression, star-convex priors, shape-penalty terms), never as non-differentiable RL verifiers.

Biological/topological structure is the verifier for static pathology segmentation.

Method

Policy. Use a SAM/SAM2 image-branch mask decoder as the policy π_θ . GRPO with group-relative advantages; no critic.

Composite reward.

$$R = R_{\text{seg}} + \lambda_{\text{cnt}} \cdot R_{\text{cnt}} + \lambda_{\text{sep}} \cdot R_{\text{sep}} + \lambda_{\text{topo}} \cdot R_{\text{topo}} + \lambda_{\text{match}} \cdot R_{\text{match}} + R_{\text{fmt}}$$

- **R_seg** — grounding term: overlap on the sparsely-labeled pixels, or Dice on the small fully-annotated anchor set. Keeps masks attached to real evidence.
- **R_cnt** — instance-count agreement: $-|\#\text{connected-components}(\text{pred}) - c|$, where c is the point-annotation count. *Non-differentiable* (thresholding + CC labeling).
- **R_sep** — instance separation: reward a 1:1 match between decomposed instances and point markers; penalize a single predicted component covering ≥ 2 markers (the merged-nuclei failure). *Non-differentiable*.
- **R_topo** — topological correctness against the known structural prior: $\beta_0 = \#\text{instances}$, $\beta_1 = \#\text{lumina (glands)}$; score via Betti-number / Euler-characteristic match on the binarized mask. *Non-differentiable*.
- **R_match** — panoptic-style instance matching (Hungarian) $\rightarrow$ Panoptic Quality as reward on the anchor set. *Non-differentiable* (assignment problem).
- **R_fmt** — validity: well-formed, non-degenerate masks.

Label-efficiency. R_{cnt} , R_{sep} , R_{topo} derive from **point annotations plus a fixed biological prior**, not dense masks; only a small anchor set carries full masks (for R_{seg} / R_{match}). This is the mechanism by which the method trains from cheap supervision — the discrete verifiers convert weak labels into a usable signal that a pixel loss cannot.

R_{cnt} / R_{sep} / R_{topo} / R_{match} are non-differentiable by construction. There is no smooth penalty that reproduces them without softening away the discrete criterion.

Datasets

- **Nuclei (instance + class):** PanNuke, CoNSeP, MoNuSeg, Lizard. Instance masks are degraded to centroids to simulate the point-annotation regime; counts come for free.
- **Gland (topology):** GlaS, CRAG — for R_{topo} with $\beta_1 = \#\text{lumina}$.

Baselines

1. Supervised Dice/cross-entropy on the same weak labels (lower bound).

2. Supervised on full masks (upper bound, for reference).
3. HoVer-Net / StarDist / CellPose at matched supervision (domain SOTA).
4. **A differentiable soft-surrogate of the identical structural objectives** (soft-cIDice + a persistent-homology Betti term + a soft density-count term), trained as a loss on the same weak labels.

Ablation plan (one variable per rung)

- **Rung 0:** anchor-only supervised fine-tune (grounding sanity check).
- **Rung 1:** + R_cnt. Does count supervision alone recover instances the pixel loss merges?
- **Rung 2:** + R_sep. Isolate the clustered-nuclei failure.
- **Rung 3:** + R_topo (glands). Isolate lumen topology.
- **Rung 4:** + R_match. Full reward.
- **Go/no-go:** Rung 4 vs. Baseline 4 at matched supervision. If the discrete-reward policy wins, the surrogate gap is real and RL closes it; if it only matches, the honest finding is that surrogates already capture the signal and the contribution is label-efficiency, not non-differentiability.

References

RL and policy optimization

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Reading roadmap

Stage 0 — RL & policy optimization — PPO [1] → GRPO [2] → DeepSeek-R1 [3] → DAPO [4].

Stage 1 — Segmentation-as-RLVR: Seg-Zero [5] and Seg-R1 [6]. New and worth close reading as prior art to differentiate from: LENS [7], MedGround-R1 [9], MedSAM-Agent [10]. The differentiation line: these do grounding / interactive clicking with overlap rewards; none use non-differentiable structural verifiers for dense instance masks.

Stage 2 — Segmentation backbones & SAM: U-Net [12], nnU-Net [13], SAM [14], SAM 2 [15].

Stage 3 — Nuclei & gland instance segmentation: HoVer-Net [16] first — the horizontal/vertical-distance trick for separating clustered nuclei is exactly the failure R_{sep} targets. Then StarDist [17], Cellpose [18], distance-map regression [19], DCAN [20] for glands. Panoptic Quality [21]

Stage 4 — Weak / point supervision: SC-Net [22] and the point-annotation lineage [23]

Stage 5 — Topological & structural objectives: Hu 2019 [24] and Clough 2019 [25] to see how persistent homology is used to make Betti numbers differentiable; cIDice [26] for the soft-skeleton surrogate; Stucki 2023 [27] and Berger 2024 [28] for the current frontier.

Stage 6 — RL in pathology / WSI — OPTIONAL/FAST (positioning + fallback variant). RLogist [30], PEAN [31], Qaiser & Rajpoot [29] for the “RL-in-pathology is navigation/classification” framing; stateless actor-critic [32] as the nearest instance-segmentation-with-priors RL prior art.